JOSIAH STEVE SAM

AI Engineer | Agentic & Generative AI Systems | Chennai, India (Open to Relocation)

+91 7373077757 · josiahsteve29@gmail.com · LinkedIn · GitHub

PROFESSIONAL SUMMARY

AI Engineer with 2 years shipping production agentic and RAG systems for banking, SaaS, and EdTech clients. Own end-to-end architecture for agentic workflows, hybrid retrieval, evaluation pipelines, and GPU-accelerated inference using LangGraph, FastAPI, and AWS. Track record of measurable gains in accuracy, latency, and reliability, plus client-facing technical scoping and mentorship.

TECHNICAL SKILLS

Agentic & GenAI: LangGraph, LangChain, Tool-Calling, RAG (Hybrid Retrieval, Reranking), Prompt Engineering, LLM Evaluation & Guardrails, BGE Embeddings, Transformers, NLP

LLM Providers & ML Frameworks: OpenAI API, LiteLLM, Meta Llama, HuggingFace, PyTorch, Scikit-learn, Pandas, NumPy

Backend & Systems: FastAPI, Django, Microservices, REST APIs, SSE Streaming, OAuth2/JWT, Nginx, Docker, Linux (Ubuntu/Rocky)

Data & Vector Search: PostgreSQL (RLS), MongoDB, Redis, Qdrant, Supabase

Cloud & MLOps: AWS (EC2, S3), GPU-Optimized Inference, Model Serving, CI/CD, Langfuse Tracing, Logging & Monitoring

Languages & Tools: Python, SQL, JavaScript, Git, VS Code, Cursor

PROFESSIONAL EXPERIENCE

Credentech Solutions · Software Engineer (AI) *Apr 2026 – Present*
Chennai, India

- Own architecture decisions across retrieval, inference, and multi-tenant trade-offs, taking AI services from design to production; standardized design patterns now used across 2+ production services.
- Design and build production LLM applications end to end: RAG pipelines (ingestion, chunking, embeddings, hybrid retrieval, reranking), model gateways, evaluation, and guardrails.
- Architect agentic workflows with tool calling, multi-step reasoning, and structured outputs using LangChain and LangGraph.
- Lead client-facing technical scoping for AI deployment engagements and mentor an incoming engineer on RAG pipeline design and production debugging.

Credentech Solutions · Junior Software Engineer *Feb 2025 – Apr 2026*
Chennai, India

- Progressed from feature development to independently building end-to-end AI and backend solutions across 4 client projects in banking, SaaS, food & beverage, and travel.
- Developed OCR and document-intelligence pipelines for identity, financial, and logistics documents across multilingual use cases, including GPU-accelerated inference with automated failover.
- Built REST APIs and microservices integrating OCR, LLM-based extraction, validation, and structured JSON output with FastAPI, PostgreSQL, and MongoDB.
- Secured multi-tenant APIs with OAuth2 client credentials and JWT, and maintained a PostgreSQL audit trail that cut manual audit effort by ~40%.

Quelit · Data Science Intern *Jul 2024 – Dec 2024*
Chennai, India

- Built a persona-consistent chatbot (OpenAI API, Streamlit, MongoDB) with multi-turn conversation storage for 20+ users.
- Designed pipelines to capture and analyze user interactions, applying NLP (sentiment analysis, text processing) to 2,000+ unstructured messages to surface persona and behavioral insights.

- Built and compared Scikit-learn regression models (R^2, RMSE, MAE) in an end-to-end pipeline predicting student academic performance.

PROJECTS

Credentra Agents: AI Agent Builder Platform (B2B SaaS) *Credentech Solutions*
FastAPI, Next.js, PostgreSQL, Qdrant, LiteLLM, NeMo Guardrails, Langfuse, Crawl4AI, Docker

- Launched a B2B SaaS where any business builds and publishes its own AI agents on 3 channels (website widget, chat page, WhatsApp) by decoupling channel adapters from one conversation runtime.
- Achieved ~90% grounded-answer accuracy at ~2s latency with hybrid retrieval: 4-branch Qdrant search (dense, FAQ, sparse, exact-code) fused via RRF, PostgreSQL authorization, and reranking.
- Raised retrieval hit@1 from 0.77 to 0.95 on a live evaluation set by adding a cross-encoder reranker, with an offline evaluation harness gating every versioned retrieval-profile change.
- Cut unsafe/off-policy responses by ~30% by routing every model call through an 8-step gateway (rate limits, NeMo guardrails, fallbacks, tracing) and a citation fence that fails closed.
- Cut new-tenant setup from days to ~1 hour with self-serve ingestion of text, FAQs, files (PDF, DOCX, XLSX), and websites, using versioned revisions that go live only after an automated knowledge check.
- Reduced critical-issue response time by ~40% and time-to-identify content gaps by ~50% by building human takeover with a live inbox and a report that groups unanswered questions by meaning.
- Isolated tenant data across 22 tables with PostgreSQL row-level security, workspace-filtered vector search, and role-based access control (3 roles, 13 permissions).

OCR Pipeline – APB Bank, Laos *Credentech Solutions*
FastAPI, NVIDIA L4 GPU, Nginx, PostgreSQL, OAuth2/JWT

- Reached ~95% field-extraction accuracy on 500+ KYC documents/day (passports, national IDs, family cards) by pairing automated image correction with GPU-accelerated field extraction.
- Sustained ~99% uptime under production load by architecting fault-tolerant multi-node inference with automatic GPU-to-CPU failover behind FastAPI microservices and Nginx.

OCR Invoice Automation – Taco Bell Thailand *Credentech Solutions*
Typhoon OCR, LLMs, FastAPI, React, Vite, PostgreSQL

- Automated mixed Thai/English invoice processing at ~95% per-document accuracy across 5 templates by combining open-source OCR with rule-based field mapping.
- Kept financial data on the client's own infrastructure by deploying the full stack on-premises, with a review dashboard and Excel export for the finance team.

AI-Powered CBSE Learning Platform – Co-Founding Engineer *Freelance · 2026*
FastAPI, React, Next.js, Supabase, RAG, LLMs, Render

- Built RAG-based AI subject tutors personalized per student class and subject, grounded in curriculum retrieval over CBSE content.
- Designed a hybrid scoring engine tracking 25 CBSE-aligned learning parameters (Bloom's Taxonomy, metacognition), fusing rule-based scoring with an LLM personalization layer.
- Cut estimated manual grading time by ~50% by building an LLM-driven OCR and evaluation engine for rubric-based answer-sheet grading.
- Structured a CBSE question bank from thousands of past-year questions and shipped the 2-month MVP end to end on Render, iterating on live demo-user feedback.

EDUCATION

Vellore Institute of Technology, *Vellore, India* *Sept 2020 – May 2025*
Integrated M.Sc., Computational Statistics and Data Analytics · CGPA: 7.64